

Part I: Repository Analysis

1 Problem summary and basic definitions

An open question in mechanistic interpretability is how the computational capacity of a transformer depends on its number of attention heads. For this purpose we will study Boolean functions

$$f : \{0, 1\}^n \longrightarrow \{0, 1\}$$

computed by a single attention layer. For this transformer, we will take the input sequence to be

$$(x_1, \dots, x_n, =),$$

where $=$ represents a fixed query token (called the readout token), and the final classifier reads only the residual stream at that readout token. A more explicit definition of the single attention layer transformer is given in Appendix [A](#).

To determine whether this model can represent a given Boolean function f , we only need to ask whether the scalar readout score $S(x)$ can separate outputs to be labelled 1 from those to be labelled 0:

$$S(x) = c + \sum_{h=1}^H \langle w, y^{(h)}(x) \rangle, \quad f(x) = 1 \iff S(x) > 0, \quad (1)$$

where $y^{(h)}(x) \in \mathbb{R}^{d_{\text{model}}}$ is the contribution of attention head h at the fixed query/readout token, $w \in \mathbb{R}^{d_{\text{model}}}$ is the readout vector, $\langle u, v \rangle = u^\top v$ is the usual Euclidean inner product, and $c \in \mathbb{R}$ absorbs the fixed query residual and the readout threshold.¹

We define the head complexity of f as

$$H^*(f) = \min \{H : f \text{ can be computed by an } H\text{-head model}\}.$$

Our problem then is to find the smallest H for which S is positive on $f^{-1}(\{1\}) := \{x : f(x) = 1\}$ and negative on $f^{-1}(\{0\}) := \{x : f(x) = 0\}$.

1.1 Attention atoms and scalar normal form

We define the attention atom as the scalar contribution of head h to the readout,

$$\phi_h(x) = \langle w, y^{(h)}(x) \rangle = \frac{\sum_{j \in [n] \cup \{=\}} \exp(\langle q^{(h)}, k_j^{(h)}(x) \rangle) \langle w, W_O^{(h)} v_j^{(h)}(x) \rangle}{\sum_{j \in [n] \cup \{=\}} \exp(\langle q^{(h)}, k_j^{(h)}(x) \rangle)}. \quad (2)$$

where we have used the expressions in Appendix [A](#) to write ϕ_h in terms of $q^{(h)}, k_j^{(h)}(x), v_j^{(h)}(x) \in \mathbb{R}^{d_{\text{head}}}$, while $W_O^{(h)} v_j^{(h)}(x), w \in \mathbb{R}^{d_{\text{model}}}$. Note that $q^{(h)}$ is independent of x because the query token is fixed. Here

$$q^{(h)} = W_Q^{(h)} u_=:, \quad k_j^{(h)}(x) = W_K^{(h)} u_j(x), \quad v_j^{(h)}(x) = W_V^{(h)} u_j(x),$$

¹In Eq. (1) we have adopted a convention where $S(x) > 0$ corresponds to $f(x) = 1$.

with the embeddings $u_j(x)$.

From Eq. (1), the score $S(x)$ can be written as a sum over H atoms,

$$S(x) = c + \sum_{h=1}^H \phi_h(x).$$

At first sight, an attention atom contains a discouraging number of moving parts. On binary inputs, however, we will see that its dependence on x simplifies dramatically.

Proposition 1 (Affine form of an attention atom). *For every head h , there exist affine functions $N_h, D_h : \mathbb{R}^n \rightarrow \mathbb{R}$ such that, for $x \in \{0, 1\}^n$,*

$$\phi_h(x) = \frac{N_h(x)}{D_h(x)}, \quad D_h(x) > 0.$$

Proof.

Using the definition of the atom in Eq. (2), we can fix a particular head and suppress the index h . Since the query token is always fixed, its query vector q is independent of x . At input position i , we can write the embedded token as

$$u_i(x) = e_0 + p_i + x_i(e_1 - e_0).$$

The corresponding attention logit is

$$\langle q, W_K u_i(x) \rangle = \langle q, W_K(e_0 + p_i) \rangle + x_i \langle q, W_K(e_1 - e_0) \rangle.$$

Defining

$$\rho_i := \exp(\langle q, W_K(e_0 + p_i) \rangle) > 0, \quad \alpha := \exp(\langle q, W_K(e_1 - e_0) \rangle) > 0,$$

the unnormalized attention weight at position i then takes the form

$$\exp(\langle q, W_K u_i(x) \rangle) = \rho_i \alpha^{x_i} = \rho_i (1 + (\alpha - 1)x_i),$$

where we have used that $x_i \in \{0, 1\}$ to write the exponential in a convenient form. Similarly, composing the value and output maps with the final readout gives a scalar value that is affine in x_i

$$\langle w, W_O W_V u_i(x) \rangle = m_i + \delta x_i, \quad m_i, \delta \in \mathbb{R}.$$

Putting this together we can find

$$\phi(x) = \frac{N(x)}{D(x)},$$

where

$$N(x) := \eta + \sum_{i=1}^n \rho_i (1 + (\alpha - 1)x_i) (m_i + \delta x_i) = \eta + \sum_{i=1}^n \rho_i [m_i + ((\alpha - 1)m_i + \alpha \delta)x_i],$$

$$D(x) := \gamma + \sum_{i=1}^n \rho_i \alpha^{x_i} = \gamma + \sum_{i=1}^n \rho_i [1 + (\alpha - 1)x_i],$$

where the fixed query token contributes constants $\eta \in \mathbb{R}$ and $\gamma > 0$ to the numerator and denominator, and we have used $x_i^2 = x_i$ to simplify the expressions above. Therefore N and D are affine in x . Moreover,

$$D(x) = \gamma + \sum_{i=1}^n \rho_i \alpha^{x_i} > 0, \quad x \in \{0, 1\}^n,$$

so D is strictly positive on the Boolean cube. Restoring the subscripts h and repeating for each head, the result follows. $\square$

1.2 Threshold Degree

This affine numerator–denominator form will be a foundational tool throughout the rest of this write-up. To use it, we introduce the notion of threshold degree.

A polynomial $P : \mathbb{R}^n \rightarrow \mathbb{R}$ *sign-represents* a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ if

$$P(x) > 0 \quad \text{when } f(x) = 1, \quad P(x) < 0 \quad \text{when } f(x) = 0.$$

The *threshold degree* of f is the minimum degree of any polynomial that sign-represents it:

$$\deg_{\pm}(f) := \min \{ \deg P : P \text{ sign-represents } f \}. \quad (3)$$

Given that $f(x)$ is only defined on the Boolean cube, a sign-representing polynomial need only reproduce the two output classes for $x \in \{0, 1\}^n$ and is generally not unique. Using our previous result of Proposition 1, we can now show how the head complexity H^* is bounded by the threshold degree.

Proposition 2 (Threshold degree bounds head complexity). *For every Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$,*

$$\deg_{\pm}(f) \leq H^*(f).$$

Proof.

Suppose that f is computed using H attention heads. By the affine form of an attention atom, its classification score can be written as

$$S(x) = c + \sum_{h=1}^H \frac{N_h(x)}{D_h(x)}, \quad D_h(x) > 0,$$

where each N_h and D_h is affine. We may assume that $S(x)$ is strictly positive when $f(x) = 1$ and strictly negative when $f(x) = 0$.

Define the common denominator

$$D(x) := \prod_{h=1}^H D_h(x) > 0.$$

Multiplying the score by $D(x)$ gives a polynomial

$$\begin{aligned} P(x) &:= D(x)S(x) \\ &= c \prod_{h=1}^H D_h(x) + \sum_{h=1}^H N_h(x) \prod_{\substack{j=1 \\ j \neq h}}^H D_j(x). \end{aligned}$$

Each term is a product of at most H affine functions, and hence

$$\deg P \leq H.$$

Since $D(x) > 0$, $P(x)$ and $S(x)$ have the same sign on every Boolean input. Therefore P sign-represents f . By the definition of threshold degree,

$$\deg_{\pm}(f) \leq \deg P \leq H.$$

This holds for every H -head representation of f . Taking $H = H^*(f)$, the minimum number of heads that computes f , gives

$$\deg_{\pm}(f) \leq H^*(f),$$

as required. □

Then any lower bound on the threshold degree of f immediately gives the same lower bound on its head complexity.

1.3 Walsh character and bounds on PARITY

It will be convenient to pass from the bits $x_i \in \{0, 1\}$ to sign variables

$$z_i = 1 - 2x_i \in \{-1, 1\}.$$

For every subset $S \subseteq [n]$, define the corresponding *Walsh character*

$$\chi_S(z) := \prod_{i \in S} z_i, \quad \chi_{\emptyset}(z) = 1.$$

These characters include the constant function, the individual coordinates, and their higher-order products:

$$1, \quad z_i, \quad z_i z_j, \quad z_i z_j z_k, \quad \dots$$

Because $z_i^2 = 1$ on the Boolean cube, repeated powers carry no additional information. Every polynomial restricted to $\{-1, 1\}^n$ can therefore be reduced to a multilinear polynomial in which each variable appears at most once. The 2^n possible products χ_S are linearly independent and form a complete basis for all real-valued functions on the 2^n points of the Boolean cube. Consequently, every such function has a unique expansion

$$F(z) = \sum_{S \subseteq [n]} \hat{F}(S) \chi_S(z).$$

This expansion will be incredibly useful in the rest of this analysis. We can immediately use it to fix the degree of PARITY_n .

Proposition 3 (Threshold degree of parity). *For n -bit parity,*

$$\deg_{\pm}(\text{PARITY}_n) = n.$$

Consequently,

$$H^*(\text{PARITY}_n) \geq n.$$

Proof.

In sign variables, parity is

$$q(z) = -\chi_{[n]}(z) = -\prod_{i=1}^n z_i,$$

so q itself is a degree- n sign representative and

$$\deg_{\pm}(\text{PARITY}_n) \leq n.$$

Suppose instead that some polynomial P of degree less than n sign-represents parity. On the Boolean cube it has an expansion

$$P(z) = \sum_{S \subseteq [n]} c_S \chi_S(z).$$

Using orthogonality of the Walsh characters under the uniform distribution,

$$\mathbb{E}_z[q(z)P(z)] = - \sum_{S \subseteq [n]} c_S \mathbb{E}_z[\chi_{[n]}(z)\chi_S(z)] = 0,$$

because $\chi_{[n]}$ is orthogonal to every lower-degree character χ_S , since there will always exist an uncanceled factor z_i in the product when $S \neq [n]$.

However, sign representation requires

$$q(z)P(z) > 0$$

at every Boolean input, so its uniform expectation must be strictly positive. This contradiction proves

$$\deg_{\pm}(\text{PARITY}_n) = n.$$

The general bound $\deg_{\pm}(f) \leq H^*(f)$ then gives

$$H^*(\text{PARITY}_n) \geq n.$$

□

The matching upper bound

$$H^*(\text{PARITY}_n) \leq n$$

is established constructively in Appendix B. Together with Proposition 3, this gives

$$H^*(\text{PARITY}_n) = n.$$

Part II: Symmetry and head complexity

One useful way to analyze head complexity is through symmetries of the Boolean function f . The same symmetry may be shared by a large class of functions, allowing them to be studied within a common framework.

As a motivating example, consider **complement symmetry**. We say that a Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ has *complement symmetry* if

$$f(\mathbf{1} - x) = f(x) \quad \text{for every } x \in \{0, 1\}^n.$$

This symmetry holds for functions whose output depends only on relationships between the bits, rather than on which Boolean value is labelled 0 or 1. Examples of such functions include $\text{XOR}(x_1, x_2)$ and $\text{EQUALITY}(x_1, x_2)$. This symmetry alone gives us a lower bound on the head complexity of every nonconstant function in this class.

Proposition 4. *Every nonconstant Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ with complement symmetry satisfies*

$$H^*(f) \geq 2.$$

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Proof.

Set $z_i = 1 - 2x_i$ and $q(z) = 2f(x) - 1$. Complement symmetry becomes

$$q(-z) = q(z).$$

Suppose $\deg_{\pm}(f) < 2$. Then an affine polynomial

$$P(z) = a_0 + \sum_{i=1}^n a_i z_i$$

strictly sign-represents q , so

$$q(z)P(z) > 0, \quad q(z)P(-z) > 0$$

for every $z \in \{-1, 1\}^n$. Averaging these inequalities gives

$$q(z) \left(\frac{P(z) + P(-z)}{2} \right) = q(z) a_0 > 0, \quad \text{for every } z \in \{-1, 1\}^n.$$

Thus q has the same sign everywhere and, because $q \in \{-1, 1\}$, is constant. This contradicts the assumption that f is nonconstant. Therefore

$$\deg_{\pm}(f) \geq 2.$$

Using $\deg_{\pm}(f) \leq H^*(f)$, we conclude that

$$H^*(f) \geq 2.$$

□

This argument extends to functions invariant under finite group actions for which the Reynolds average of every linear term vanishes. We defer this generalization to Appendix C.

1.4 Cyclic groups, invariants, and orbits

Determining the threshold degree of a Boolean function directly can require searching over arbitrary polynomials on all 2^n inputs. When the function has a symmetry, we can reduce this search to separating a much smaller set of input orbits using invariant polynomials. In this section, we will demonstrate this reduction by studying Boolean functions that are invariant under cyclic groups generated by a linear transformation g .

Let

$$G = \langle g \rangle = \{e, g, \dots, g^{M-1}\}$$

be a cyclic group of order M acting on the Boolean cube²

$$\Omega = \{-1, 1\}^n.$$

We assume that g is linear and maps Ω to itself. Such a transformation is necessarily a signed coordinate permutation, and G is a cyclic subgroup of the signed permutation group. Since g has order M ,

$$g^M = e,$$

where e is the identity.

A function $F : \Omega \rightarrow \{-1, 1\}$ is G -invariant if

$$F(g^r z) = F(z), \quad r = 0, \dots, M-1.$$

Therefore applying the transformation g does not change the value of F .

For any representative vector $z^{(a)} \in \Omega$, repeatedly applying g produces its orbit

$$O_a = \text{Orb}_G(z^{(a)}) = \{z^{(a)}, gz^{(a)}, \dots, g^{M-1}z^{(a)}\}.$$

Because $g^M = e$, the orbit contains at most M distinct inputs. Suppose the Boolean cube is divided into m distinct orbits,

$$O_1, \dots, O_m \subseteq \Omega.$$

A G -invariant function takes the same value on every input within an orbit. Therefore, instead of specifying F separately on all 2^n inputs, it is enough to specify the m values

$$F(O_1), \dots, F(O_m) \in \{-1, 1\},$$

where we use $F(O_i)$ as a shorthand for evaluating F on some representative vector $z^{(i)} \in O_i$.

To distinguish these m orbits, we can expand

$$F(z) = \sum_a c_a I_a(z), \tag{4}$$

²For this analysis, we replace the variables $x_i \in \{0, 1\}$ by the sign variables

$$z_i = 2x_i - 1 \in \{-1, 1\}.$$

We likewise encode the Boolean outputs as $\{-1, 1\}$. This affine change of variables does not alter polynomial degree or head complexity.

in a basis of G -invariant functions I_a . Like F , each invariant I_a is constant on every orbit and therefore defines an orbit-evaluation vector

$$v_a = (I_a(O_1), I_a(O_2), \dots, I_a(O_m))^T \in \mathbb{R}^m.$$

Choosing m linearly independent vectors v_a gives a basis capable of representing any assignment of values to the m orbits.

Defining

$$V_{ia} = I_a(O_i),$$

the coefficients in Eq. (4) are uniquely determined by

$$\mathbf{c} = V^{-1} (F(O_1), F(O_2), \dots, F(O_m))^T.$$

This gives an invariant representation that agrees with F exactly. To sign-represent F however, exact agreement is unnecessary. We may instead seek coefficients s_a satisfying

$$F(O_i) \sum_a s_a I_a(O_i) > 0 \quad \text{for every } O_i.$$

The coefficients $s_a = c_a$ always provide one solution, but another solution may omit some of the higher-degree invariants and therefore give a stronger upper bound on $\deg_{\pm}(F)$. A systematic procedure for finding the lowest-degree solution is given in Appendix D.

1.5 Reynolds averaging and the invariant basis

While some might worry that finding these G -invariant functions I_a would be tedious, constructing such a basis is actually straightforward. Given any function $H : \Omega \rightarrow \mathbb{R}$, define its Reynolds average by

$$(\mathcal{R}_G H)(z) = \frac{1}{M} \sum_{r=0}^{M-1} H(g^r z).$$

This function is G -invariant, since

$$(\mathcal{R}_G H)(gz) = \frac{1}{M} \sum_{r=0}^{M-1} H(g^{r+1} z) = (\mathcal{R}_G H)(z),$$

where $g^M = e$ ensures that shifting $r \mapsto r + 1$ only permutes the terms in the sum.

Proposition 5 (Completeness of the invariant features). *Let $G = \langle g \rangle$ act on $\Omega = \{-1, 1\}^n$, with g linear. Every G -invariant function $F : \Omega \rightarrow \mathbb{R}$ can be expressed as*

$$F(z) = \sum_{S \subseteq [n]} c_S I_S(z),$$

for suitable coefficients $c_S \in \mathbb{R}$. Moreover, if $\deg(F) \leq d$, then only invariants of degree at most d are required.

Proof. Recall the Walsh characters introduced in Section 1.3:

$$\chi_S(z) = \prod_{i \in S} z_i, \quad S \subseteq [n].$$

Reynolds-averaging each character gives the invariant feature

$$I_S(z) = (\mathcal{R}_G \chi_S)(z) = \frac{1}{M} \sum_{r=0}^{M-1} \chi_S(g^r z).$$

The Walsh characters form a basis for all real-valued functions on the Boolean cube, so

$$F(z) = \sum_{S \subseteq [n]} \hat{F}(S) \chi_S(z).$$

Since F is G -invariant, $\mathcal{R}_G F = F$. Therefore,

$$F(z) = (\mathcal{R}_G F)(z) = \sum_{S \subseteq [n]} \hat{F}(S) (\mathcal{R}_G \chi_S)(z) = \sum_{S \subseteq [n]} \hat{F}(S) I_S(z).$$

Hence, the I_S span the space of G -invariant functions.

It remains to check that this construction respects polynomial degree. The Walsh character χ_S is a multilinear monomial containing $|S|$ variables and therefore has degree $|S|$. Because g is linear, composition with g^r does not increase polynomial degree. Therefore,

$$\deg(I_S) = \deg(\mathcal{R}_G \chi_S) \leq \deg(\chi_S) = |S|.$$

If $\deg(F) \leq d$, then

$$\hat{F}(S) = 0 \quad \text{whenever} \quad |S| > d.$$

Therefore, only invariants I_S of degree at most d are required in the expansion of F . $\square$

The previous proposition shows that the functions I_S span every degree-bounded G -invariant polynomial on the Boolean cube Ω . To use this basis to study threshold degree, we now show that restricting to G -invariant sign-representing polynomials loses no generality: any non-invariant sign representation can be Reynolds-averaged without increasing its degree.

Proposition 6 (Invariant threshold-degree bound). *Let $G = \langle g \rangle$ act linearly on $\Omega = \{-1, 1\}^n$, and let $F : \Omega \rightarrow \{-1, 1\}$ be G -invariant. Then*

$$\deg_{\pm}(F) = \min_{\substack{P \text{ } G\text{-invariant} \\ F(z)P(z) > 0 \ \forall z \in \Omega}} \deg(P) \leq H^*(F).$$

Proof. Let P be a lowest-degree G -invariant polynomial that sign-represents F . Suppose that a not necessarily invariant polynomial w also sign-represents F , with

$$\deg(w) < \deg(P).$$

Since $F(hz) = F(z)$ for every $h \in G$,

$$F(z)(\mathcal{R}_G w)(z) = \frac{1}{|G|} \sum_{h \in G} F(hz)w(hz) > 0.$$

Thus, $\mathcal{R}_G w$ is a G -invariant polynomial that sign-represents F . By Proposition 5, Reynolds averaging does not increase polynomial degree. Therefore,

$$\deg(\mathcal{R}_G w) \leq \deg(w) < \deg(P),$$

contradicting the minimality of P . Combining this result with Proposition 2, it follows then that

$$\deg_{\pm}(F) = \deg(P) \leq H^*(F).$$

□

1.6 A four-cycle example

To demonstrate the invariant construction, consider the order-four cyclic group

$$G = \langle g \rangle = \{e, g, g^2, g^3\}, \quad g(z_1, z_2, z_3, z_4) = (z_2, z_3, z_4, z_1),$$

acting on $\Omega = \{-1, 1\}^4$. Before choosing a particular G -invariant function F , we first determine the orbit structure induced by G .

For compactness, we represent each input by a string of signs. Repeatedly applying g partitions the 16 inputs into $m = 6$ orbits:

Orbit	Representative	Elements
O_0	----	{----}
O_1	+-++	{+-++, -++-, -++-, -++-}
O_{adj}	++--	{++--, -++-, -++-, -++-}
O_{opp}	+-+-	{+-+-, -+-+}
O_3	----	{----, +---, +---, +---}
O_4	----	{----}

The subscripts on O_0, O_1, O_3, O_4 give the Hamming weight of any input z in the corresponding orbit, where we define the Hamming weight as the number of positive coordinates. Hamming weight uniquely determines the orbit except when two coordinates are positive. In that case, the inputs split into the adjacent orbit O_{adj} and the opposite orbit O_{opp} , distinguished by their arrangement around the cycle.

We next construct invariant functions by Reynolds-averaging the Walsh characters. For example, averaging

$$\chi_{\{1,2\}}(z) = z_1 z_2$$

gives

$$\begin{aligned} I_{\text{adj}}(z) &= (\mathcal{R}_G \chi_{\{1,2\}})(z) \\ &= \frac{1}{4} (z_1 z_2 + z_2 z_3 + z_3 z_4 + z_4 z_1). \end{aligned}$$

Applying the same procedure to the remaining Walsh characters gives the following basis of

invariant functions:

$$\begin{aligned}
I_{\emptyset}(z) &= 1, & I_1(z) &= \frac{1}{4} \sum_{i=1}^4 z_i, \\
I_{\text{adj}}(z) &= \frac{1}{4} \sum_{i=1}^4 z_i z_{i+1}, & I_{\text{opp}}(z) &= \frac{1}{2} (z_1 z_3 + z_2 z_4), \\
I_3(z) &= \frac{1}{4} \sum_{i=1}^4 z_i z_{i+1} z_{i+2}, & I_4(z) &= z_1 z_2 z_3 z_4.
\end{aligned}$$

Each invariant I_a is constant on each orbit and therefore defines a vector

$$v_a = (I_a(O_0), I_a(O_1), I_a(O_{\text{adj}}), I_a(O_{\text{opp}}), I_a(O_3), I_a(O_4))^{\text{T}} \in \mathbb{R}^6.$$

These six vectors are linearly independent, so every G -invariant function can be written uniquely as a linear combination of the six invariants. A G -invariant Boolean function is equivalently specified by assigning either $+1$ or -1 to each orbit. Since the $m = 6$ orbit labels can be chosen independently, there are

$$2^m = 2^6 = 64$$

distinct G -invariant Boolean functions.

We'll proceed with the rest of this analysis by studying one particular function

$$F(z) = \begin{cases} +1, & \text{if there exists } i \in \{1, 2, 3, 4\} \text{ such that } z_i = z_{i+1} = +1, \\ -1, & \text{otherwise,} \end{cases} \quad z_5 := z_1,$$

which checks if two adjacent bits z_i and z_{i+1} have value $+1$. We can see that this function will naturally imply,

$$F(O_{\text{adj}}) = F(O_3) = F(O_4) = +1,$$

while

$$F(O_0) = F(O_1) = F(O_{\text{opp}}) = -1.$$

The function is G -invariant because a cyclic shift maps adjacent pairs to adjacent pairs.

Following the orbit-separation procedure outlined in Appendix D, we begin with the lowest-degree nonconstant invariant I_1 and ask whether it is sufficient to separate the orbit labels. Each orbit is mapped to the point

$$p_i = I_1(O_i) \in \mathbb{R},$$

which we color according to $F(O_i)$. Since

$$I_1(z) = \frac{1}{4} \sum_{i=1}^4 z_i,$$

the orbits O_{adj} and O_{opp} have the same coordinate $I_1 = 0$ but opposite colors, so they cannot be separated using I_1 alone.

To lift this degeneracy, we introduce the degree-two invariant I_{opp} and map each orbit to

$$p_i = (I_1(O_i), I_{\text{opp}}(O_i)) \in \mathbb{R}^2,$$

whose values are given in Table 1 and a visualization of the orbits is shown in Fig. 1.

Orbit	$I_1(O_i)$	$I_{\text{opp}}(O_i)$	$F(O_i)$
O_0	-1	1	-1
O_1	$-\frac{1}{2}$	0	-1
O_{adj}	0	-1	+1
O_{opp}	0	1	-1
O_3	$\frac{1}{2}$	0	+1
O_4	1	1	+1

Table 1: Values of I_1 and I_{opp} on each orbit. The final column gives the Boolean label $F(O_i)$.

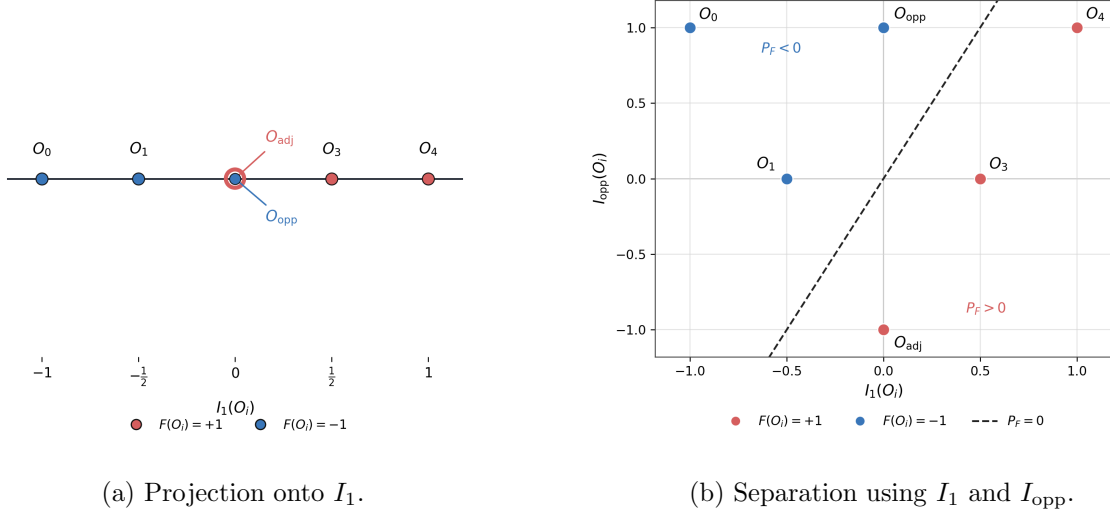


Figure 1: Orbit separation using invariant features. Colors indicate the value of $F(O_i)$.

Since

$$I_{\text{opp}}(O_{\text{adj}}) = -1, \quad I_{\text{opp}}(O_{\text{opp}}) = +1,$$

the two orbits now correspond to distinct points, and all positive and negative orbits are separated by the line

$$I_1 - \frac{1}{2}I_{\text{opp}} = 0.$$

We can immediately conclude that for this function F

$$\deg_{\pm}(F) = \deg(I_{\text{opp}}) = 2, \quad H^*(F) \geq 2.$$

2 Summary and conclusion

We developed a symmetry-based method for studying the threshold degree of Boolean functions and obtaining lower bounds on their head complexity. For a function invariant under a linear cyclic group $G = \langle g \rangle$, the group action partitions the 2^n Boolean inputs into m orbits, reducing the original problem to assigning and separating m orbit labels.

Reynolds-averaging the Walsh characters provides a complete, degree-controlled basis of invariant features. In this feature space, each orbit becomes a point and each G -invariant Boolean function becomes a coloring of these points, so finding its threshold degree reduces to finding the lowest-degree invariant features that linearly separate the two colors. Because the orbit structure and invariant basis depend only on G , the same construction can be reused for the entire class of 2^m G -invariant Boolean functions.

The four-cycle example illustrated how this description also makes the result interpretable: the degree-one invariant distinguishes Hamming-weight levels, while a degree-two invariant resolves the remaining distinction between adjacent and opposite configurations. Reynolds averaging shows that allowing non-invariant polynomials cannot lower the threshold degree. The lowest-degree invariant separator therefore gives the exact value of $\deg_{\pm}(F)$ and the lower bound

$$\deg_{\pm}(F) \leq H^*(F).$$

Lean-verified proofs of the complement-symmetry lower bound and its zero-Reynolds-average generalization are available [here](#) and [here](#), respectively.

A Attention model

For an input $x = (x_1, \dots, x_n) \in \{0, 1\}^n$, the model receives the sequence

$$(x_1, \dots, x_n, =).$$

The final symbol is a fixed query token, and only its residual stream is used by the classifier.

Let d_{model} and d_{head} denote the model and head dimensions. Choose token embeddings

$$e_0, e_1, e_{=} \in \mathbb{R}^{d_{\text{model}}}$$

and positional embeddings

$$p_1, \dots, p_n, p_{=} \in \mathbb{R}^{d_{\text{model}}}.$$

The embedded inputs are

$$u_i(x) = e_{x_i} + p_i, \quad u_{=} = e_{=} + p_{=}.$$

In particular, $u_{=}$ is independent of x .

For each head $h \in \{1, \dots, H\}$, choose matrices

$$W_Q^{(h)}, W_K^{(h)}, W_V^{(h)} \in \mathbb{R}^{d_{\text{head}} \times d_{\text{model}}}, \quad W_O^{(h)} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}.$$

At the query position, define

$$q^{(h)} = W_Q^{(h)} u_{=}, \quad k_j^{(h)}(x) = W_K^{(h)} u_j(x), \quad v_j^{(h)}(x) = W_V^{(h)} u_j(x),$$

where $j \in \{1, \dots, n, =\}$. The attention weights are

$$\alpha_j^{(h)}(x) = \frac{\exp\left(\left\langle q^{(h)}, k_j^{(h)}(x) \right\rangle\right)}{\sum_{\ell \in \{1, \dots, n, =\}} \exp\left(\left\langle q^{(h)}, k_\ell^{(h)}(x) \right\rangle\right)}.$$

The contribution of head h to the query residual is

$$y^{(h)}(x) = W_O^{(h)} \sum_{j \in \{1, \dots, n, =\}} \alpha_j^{(h)}(x) v_j^{(h)}(x).$$

The final query representation is

$$r(x) = u_{=} + \sum_{h=1}^H y^{(h)}(x).$$

For a readout vector $w \in \mathbb{R}^{d_{\text{model}}}$ and threshold $\tau \in \mathbb{R}$, the scalar score is

$$S(x) = \langle w, r(x) \rangle - \tau,$$

and the model computes f when

$$f(x) = 1 \quad \Longleftrightarrow \quad S(x) > 0.$$

The architecture contains no MLP, layer normalization, dropout, or additional attention layers. Since the fixed query token is placed last, the formulas are compatible with the usual causal-mask convention.

B An explicit n -head construction for parity

In Proposition 1, we showed that each attention head contributes one atom ϕ_h , so an H -head model has score

$$S(x) = c + \sum_{h=1}^H \phi_h(x).$$

We now construct n such atoms whose sum represents parity.

Let

$$k(x) = \sum_{i=1}^n x_i$$

be the Hamming weight of x , which counts the number of coordinates equal to 1. Choosing distinct parameters

$$\alpha_1, \dots, \alpha_n > 1,$$

for each $h = 1, \dots, n$, we can choose a head whose atom is

$$\phi_h(x) = \beta_h g_h(k(x)), \quad g_h(k) = \frac{\alpha_h k}{n + 1 + (\alpha_h - 1)k},$$

where β_h will be chosen below. Such an atom is obtained by assigning weight α_h and value 1 to every 1-token, while every 0-token and the query token have weight 1 and value 0. On an input containing k ones, the numerator is therefore $\alpha_h k$, while the denominator is

$$(n - k) + \alpha_h k + 1 = n + 1 + (\alpha_h - 1)k.$$

To show that these atoms can reproduce any function of the Hamming weight, we first show that the $n + 1$ functions

$$1, g_1, \dots, g_n$$

are linearly independent on $\{0, \dots, n\}$. Suppose

$$c_0 + \sum_{h=1}^n c_h g_h(k) = 0$$

for every $k = 0, \dots, n$. Setting $k = 0$ gives $c_0 = 0$. For $k = 1, \dots, n$, dividing by k and defining

$$r_h = \frac{n + 1}{\alpha_h - 1}, \quad d_h = \frac{c_h \alpha_h}{\alpha_h - 1},$$

gives

$$\sum_{h=1}^n \frac{d_h}{k + r_h} = 0.$$

Multiplying by $\prod_h (k + r_h)$ produces a polynomial of degree at most $n - 1$ that vanishes at the n distinct points $k = 1, \dots, n$. It must therefore vanish identically. Evaluating it at $k = -r_h$ shows that $d_h = 0$, and therefore $c_h = 0$, for every h .

As such $1, g_1, \dots, g_n$ form a basis for all functions of the Hamming weight. We may then choose coefficients $\beta_0, \beta_1, \dots, \beta_n$ satisfying

$$\beta_0 + \sum_{h=1}^n \beta_h g_h(k) = \begin{cases} +1, & k \text{ is odd,} \\ -1, & k \text{ is even,} \end{cases} \quad k = 0, \dots, n.$$

The resulting n -head score

$$S(x) = \beta_0 + \sum_{h=1}^n \phi_h(x)$$

computes parity, proving

$$H^*(\text{PARITY}_n) \leq n.$$

Together with the threshold-degree lower bound from Proposition 3, this gives

$$H^*(\text{PARITY}_n) = n.$$

C Lower bound on H^* from vanishing Reynolds average

Let G be a finite group acting linearly on $\mathbb{R}^n$. Its *Reynolds operator* is the group average

$$\mathcal{R}_G := \frac{1}{|G|} \sum_{g \in G} g.$$

For any $z \in \mathbb{R}^n$, the vector $\mathcal{R}_G z$ is the average of its images under G . We say that the action has *zero Reynolds average* when

$$\mathcal{R}_G = 0.$$

Equivalently, the vectors in every G -orbit have average zero.

Proposition 7 (Zero Reynolds average forces at least two heads). *Let G be a finite group acting linearly on $\mathbb{R}^n$ and preserving $\{-1, 1\}^n$. Let $q : \{-1, 1\}^n \rightarrow \{-1, 1\}$ be nonconstant and G -invariant:*

$$q(gz) = q(z), \quad g \in G, \quad z \in \{-1, 1\}^n.$$

If $\mathcal{R}_G = 0$, then

$$\deg_{\pm}(q) \geq 2, \quad \text{and consequently} \quad H^*(q) \geq 2.$$

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Proof.

Suppose, toward a contradiction, that $\deg_{\pm}(q) \leq 1$. Then an affine polynomial

$$P(z) = a_0 + \langle a, z \rangle$$

strictly sign-represents q :

$$q(z)P(z) > 0, \quad z \in \{-1, 1\}^n.$$

Because G preserves the Boolean cube, the same inequality holds at every transformed input gz . Using $q(gz) = q(z)$, we obtain

$$q(z)P(gz) > 0, \quad g \in G.$$

Averaging these positive quantities over G gives

$$q(z) \frac{1}{|G|} \sum_{g \in G} P(gz) > 0.$$

However,

$$\begin{aligned} \frac{1}{|G|} \sum_{g \in G} P(gz) &= a_0 + \left\langle a, \frac{1}{|G|} \sum_{g \in G} gz \right\rangle \\ &= a_0 + \langle a, \mathcal{R}_G z \rangle = a_0. \end{aligned}$$

We therefore have

$$q(z)a_0 > 0, \quad z \in \{-1, 1\}^n.$$

Since a_0 is constant and $q(z) \in \{-1, 1\}$, this forces q to have the same value everywhere, contradicting nonconstancy. Hence $\deg_{\pm}(q) \geq 2$, and the general bound $\deg_{\pm}(q) \leq H^*(q)$ gives $H^*(q) \geq 2$. $\square$

D Finding the lowest-degree invariant separator

In Section 1.5, we showed that a G -invariant Boolean function $F : \Omega \rightarrow \{-1, 1\}$ has an exact expansion

$$F(z) = \sum_a c_a I_a(z).$$

This immediately gives a sign-representing polynomial, but its degree need not be minimal: a lower-degree polynomial may have the same signs as F on the Boolean cube without reproducing its values exactly. We now give a simple procedure for finding the lowest possible degree.

To begin, we fix a candidate degree d . By Proposition 6, if a degree- d polynomial sign-represents F , then there is also a G -invariant polynomial of degree at most d that does so. By Proposition 5, every such polynomial can be written as

$$P_{\mathbf{s}}(z) = \sum_{a=1}^{k_d} s_a I_a(z),$$

where $I_1, \dots, I_{k_d}$ form a basis for the G -invariant polynomials of degree at most d .

Since both F and $P_{\mathbf{s}}$ are constant on each orbit, it is enough to impose the sign-representation condition on the m orbits:

$$F(O_i)P_{\mathbf{s}}(O_i) = F(O_i) \sum_{a=1}^{k_d} s_a I_a(O_i) > 0, \quad i = 1, \dots, m.$$

Define

$$A_{ia}^{(d)} = F(O_i)I_a(O_i).$$

The search for a degree- d sign representation is therefore the problem of finding coefficients $\mathbf{s}$ satisfying

$$A^{(d)}\mathbf{s} > \mathbf{0}.$$

Finding such coefficients $\mathbf{s}$ is a standard linear-programming feasibility problem.³ Starting from $d = 0$, we increase d until the problem becomes feasible. The first feasible degree is therefore

$$\deg_{\pm}(F) = \min \left\{ d : \exists \mathbf{s} \in \mathbb{R}^{k_d} \text{ such that } A^{(d)}\mathbf{s} > \mathbf{0} \right\}.$$

³To place the strict inequalities in standard form, let $\gamma = \min_i (A^{(d)}\mathbf{s})_i > 0$. Then $\tilde{\mathbf{s}} = \mathbf{s}/\gamma$ satisfies $A^{(d)}\tilde{\mathbf{s}} \geq \mathbf{1}$. Conversely, any solution of this non-strict system satisfies the original strict inequalities.